

Futures

Mechanics & Uses

Explained so simply a 10-year-old gets it

A binding promise to trade later at a price fixed today — how the leverage, daily settlement, and fair-value maths really work, and what futures are for.

Stock Market Mastery Roadmap • Advanced Level
Educational only — not investment, trading, or tax advice.

Welcome to the **Advanced Level!** We start with **futures** — a **promise to trade later at a price fixed today**. Imagine promising your friend in January: "I'll buy your cricket bat for ₹500 in March, no matter what it costs then." That handshake, made official and tradable, is a **futures contract**. Since you already trade F&O, this chapter sharpens the *why* behind the *what*.

What a futures contract actually is

Unlike an option (a *right*), a future is an **obligation** — both sides *must* honour it at expiry. Key parts:

- **Underlying** — *what* the promise is on: an **index** (Nifty, Bank Nifty) or a **single stock**.
- **Lot size** — you trade in **fixed bundles**, not single units (e.g., one Nifty lot = a set number of units).
- **Expiry** — the **deadline** when the promise must be settled.
- **Tick size** — the **smallest price step** the contract can move in.

Margin: a deposit, not the full price

You don't pay the whole value of a future — just a **security deposit** called **margin**. That's what makes futures **leveraged**.

- **SPAN + Exposure margin** — the deposit the exchange demands to cover risk.
- **MTM (Mark-to-Market)** — profit/loss is **settled every single day**, not just at the end.
- **Margin call** — if losses eat your deposit, the broker asks for **more money** — or closes your position.

MTM: LEVERAGE CUTS BOTH WAYS EVERY DAY

Because of **daily MTM**, if the market moves against you, money leaves your account **that same evening** — you can't just "hold and hope" like with a share. Leverage magnifies your move on the **whole lot** while you've only put down a small margin — so gains *and* losses arrive fast and daily. Respect the size.

Fair value: the cost-of-carry idea

Why does a future's price differ from today's **spot** (current) price? Because of the **cost of carrying** the position until expiry (mainly interest). This gives some useful words:

Word	Meaning	Simple picture
Basis	future price – spot price	the gap between "later" and "now"
Contango	future higher than spot	normal — paying to carry
Backwardation	future lower than spot	unusual — often signals stress/demand

THE GAP ALWAYS CLOSES AT THE FINISH LINE

As expiry nears, the future and spot prices **converge** — they *must* meet, because on expiry day the "later" is "now." Watching how the **basis** shrinks toward expiry is a neat way to read what big players expect.

Rollover: moving the promise forward

Contracts expire, but a trader may want to **keep** the position. **Rollover** = closing the expiring contract and reopening it in the **next month's** series.

ROLLOVER DATA = A SENTIMENT CLUE

How much of the market **rolls over** to next month (rollover %) and the **Open Interest (OI)** tell you whether big players are **staying in or leaving** — a peek at the crowd's conviction. High rollovers with rising OI often mean traders are **committed** to the current view.

What futures are used for

-  **Directional trading** — a clean bet on **up or down**, with no option "time decay" nibbling at you.
-  **Hedging** — protect a portfolio: if you own stocks and fear a fall, a short index future acts like **insurance** that pays when the market drops.
-  **Spreads** — calendar or basis spreads that trade the *gap* between contracts rather than pure direction.

Index vs single-stock futures

- **Index futures** (Nifty, Bank Nifty, FinNifty) — a bet on the **whole basket**; smoother, very liquid.
- **Single-stock futures** — a bet on **one company**; can move sharply on company-specific news.

Key ideas to remember

- **Futures are linear and leveraged.**  You gain or lose the **full move on the whole lot**, even though your margin is far smaller than the contract's **notional** value.
- **MTM settles daily** — leverage cuts both ways **every single day**, not just at expiry.
- **Basis & rollover reveal expectations** — they quietly show what **large participants** are thinking.

PRACTICAL EXERCISES

-  **Compute the notional value and required margin** for one Nifty future, then work out your **effective leverage** (notional ÷ margin).
-  **Track the basis** (future – spot) for one index across a **week** and watch how it **converges** into expiry.
-  **Read the monthly rollover % and OI** for one stock and interpret the **sentiment** (committed or exiting?).

REAL-WORLD APPLICATION

-  Futures are the **cleanest tool to hedge overnight/positional risk** and to express a **pure directional view without option decay**. When you just want "long or short the market," a future is often the simplest, most honest instrument.

ONE-LINE SUMMARY

A future is a **binding promise to trade later at a price set today**, bought with a small **margin** (so it's **leveraged**) and settled **daily via MTM**. Its price differs from spot by the **cost of carry** (basis → contango/backwardation) and **converges** at expiry. Use it to **bet direction cleanly, hedge a portfolio, or trade spreads** — and always size for the **whole lot**, not the margin.